

Supply Chain Performance Analysis

Inventory Optimization, Supplier Evaluation, and Transportation Performance

Objective

This project analyzes a public supply chain dataset containing 100 stock keeping units (SKUs) to evaluate inventory performance, supplier reliability, transportation efficiency, and product profitability. The objective was to analyze supply chain performance using operational and financial metrics to identify opportunities to improve inventory planning, supplier performance, transportation efficiency, and overall decision-making.

Tools Used

- Microsoft Excel
 - Data analysis
 - Calculated metrics
 - Pivot tables
- Tableau Public
 - Interactive dashboard
 - KPI visualization
 - Data filtering

Key Performance Indicators

KPI	Value
Total Revenue	\$577,604.82
Products Sold	46,099
Average Lead Time	15.96 Days
Average Profit Margin	86.21%
Average Defect Rate	2.28%

Dataset: Publicly available supply chain dataset used for educational and portfolio purposes.

Key Findings

1. Skincare products generated approximately \$241.6K in revenue, outperforming haircare (\$174.5K) and cosmetics (\$161.5K). This indicates skincare products were the strongest revenue contributor in the dataset.
2. Supplier defect rates ranged from approximately 1.80% to 2.67%, while Supplier 1 generated the highest total revenue (about \$157.5K). This highlights the importance of evaluating suppliers using both operational performance and business impact rather than a single metric.
3. Average supplier lead times ranged from 14.3 to 17.0 days, which could influence replenishment planning and inventory availability for higher-demand products.
4. The inventory risk classification highlighted SKUs with lower inventory levels and longer lead times. These products represent the greatest potential stockout risk and should be prioritized during replenishment planning.
5. Transportation cost, lead time, and profitability varied across the dataset, demonstrating the importance of evaluating logistics decisions using both financial and operational performance metrics. Comparing shipping costs, transportation modes, and estimated profit margins provides a framework for balancing logistics costs with product profitability and service performance.

Recommendations

1. Prioritize high-revenue product categories. Since skincare products generated the highest revenue, inventory planning should prioritize maintaining adequate stock levels for these products to reduce the risk of lost sales.
2. Implement supplier scorecards that track defect rates, lead time, and revenue contribution. This allows purchasing decisions to consider both operational reliability and business value.
3. Products identified as High Inventory Risk should be reviewed to determine whether higher safety stock levels or earlier reorder points would reduce stockout risk while maintaining inventory efficiency.
4. Transportation mode selection should consider both shipping cost and lead time to balance customer service expectations with logistics expenses.
5. Continue monitoring product profitability. Use profit margin analysis alongside revenue performance when making inventory and sourcing decisions to ensure resources are focused on products that generate both strong sales and healthy returns.

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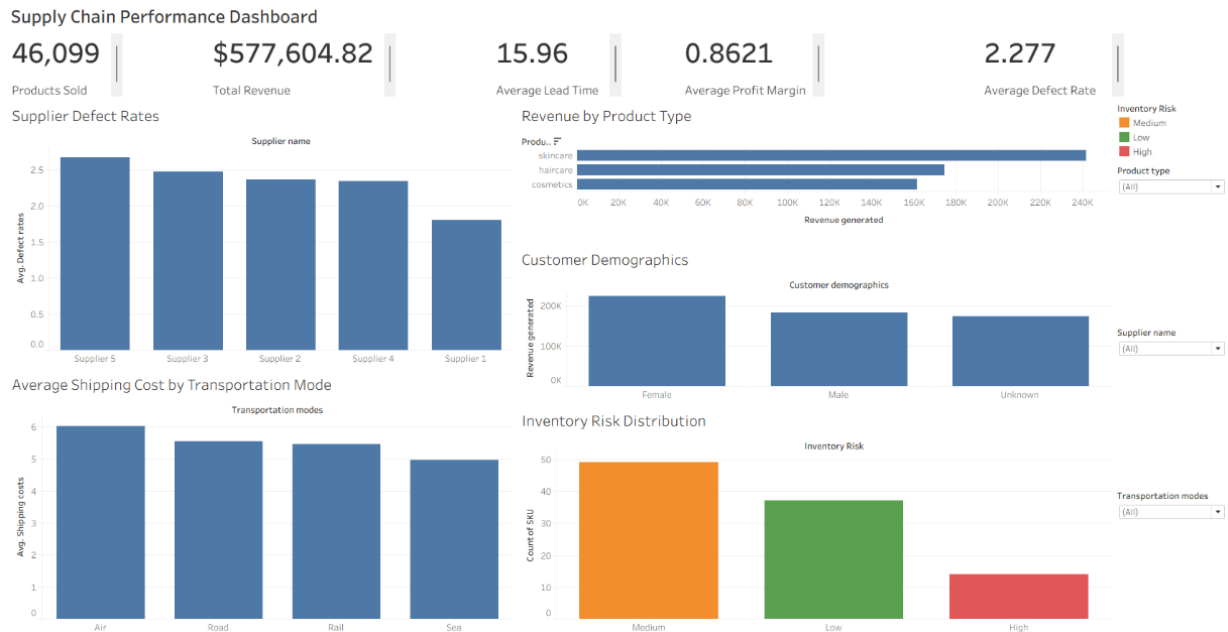


Figure 1. Interactive Tableau dashboard displaying KPIs, inventory risk, supplier performance, transportation analysis, and customer demographic trends.

Conclusion

This project demonstrates the application of Excel and Tableau to transform supply chain data into actionable business insights. The resulting analysis supports data-driven decision-making across inventory management, supplier evaluation, and logistics planning.

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